

Problem-Solving

Cognition

Definition: Cognition includes all the **mental activities** involved in:

- **Thinking**
- **Knowing**
- **Remembering**
- **Communicating**
- It's the mental toolbox that allows us to **process information**, make decisions, solve problems, and understand the world.

Concepts

- A **concept** is a **mental grouping** of similar:
 - Objects
 - Events
 - Ideas
 - People
- Concepts help us **simplify and organize** information.

Why Concepts Matter

- “Concepts keep things simple; without them, chaos would ensue.”
- Concepts help us **organize** our mental world.
- Without concepts, we'd have to treat **every object or experience as unique**. That would be overwhelming.

For example:

- The concept of “**chair**” lets us group all kinds of chairs together, even though they may look different.
- You don't have to relearn what a chair is every time you see a new one—you just match it to your concept.

How Concepts Work in Thinking

- **Fast processing:** Concepts help you make quick judgments (e.g., "That's a dog, not a cat").
- **Communication:** When someone says "fruit," you immediately picture things like apples or bananas.
- **Problem solving:** Concepts help you categorize problems and apply known solutions.

Prototypes

“For every concept, we create a prototype.”

- A **prototype** is the **best or most typical example** of a concept in your mind.
- It helps you quickly decide whether something belongs in a category.

Examples:

- For the concept “**bird**,” your prototype might be a **sparrow** or **robin**—something small and that flies.
- A **penguin** or **ostrich** doesn’t match the prototype well, so we may hesitate to label it a bird, even though it is.

Category Shifting

Over time, an item will shift toward our prototype.”

- This is a **mental shortcut** our brain uses. Once we classify something, we start thinking of it as **more similar to the prototype** than it really is.

Example:

- Suppose you meet a dog that looks and acts very differently from most dogs you know.
- Over time, you might start remembering it as **more dog-like** than it was—your brain nudges it closer to your mental “dog” prototype.

Benefits and Drawbacks

“Concepts are useful to speed up thinking, but they can lead us astray.”

Advantages:

- Speed
- Efficiency
- Quick decision-making

Disadvantages:

- **Stereotyping:** Assuming all members of a category are alike
- **Errors in judgment:** Dismissing things that don't fit our prototype
- **Miscommunication:** Different people may have different prototypes for the same concept

Cognitive Term	Definition	Example	Cognitive Skill
Concept	Mental category based on shared features	"Fruit" = sweet, edible, from plants	Categorization
Prototype	Most typical example of a concept	Apple = typical fruit	Fast recognition
Concept Shifting	Changing mental frameworks or classification rules	Switching from grouping by taste to color	Cognitive flexibility

Problem Solving in Cognition

- **Problem solving** refers to the **cognitive process of overcoming obstacles** to reach a goal when the solution is **not immediately obvious**.
- Unlike **reasoning** (which often uses general logic), **problem solving** often involves **domain-specific knowledge** and may require extended **information processing**.
- Example: Solving a physics problem, planning a sewing pattern, or playing chess — all rely on domain-specific problem solving, not just abstract logic.

Well-Defined vs. Ill-Defined Problems

- **Initial State:** Where the problem solver begins (e.g., unsolved puzzle).
- **Goal State:** The desired solution.
- **Operators:** Actions that move you from one state to another.
- **Problem Space:** All possible states you can explore to reach the goal.

Type of Problem	Characteristics	Example
Well-Defined	- Clear initial state - Clear goal - Defined rules/steps (operators)	Solving a math equation
Ill-Defined	- Unclear starting point or goal - No obvious rules - Often open-ended	Writing a novel, solving poverty

Novice vs. Expert Problem Solving

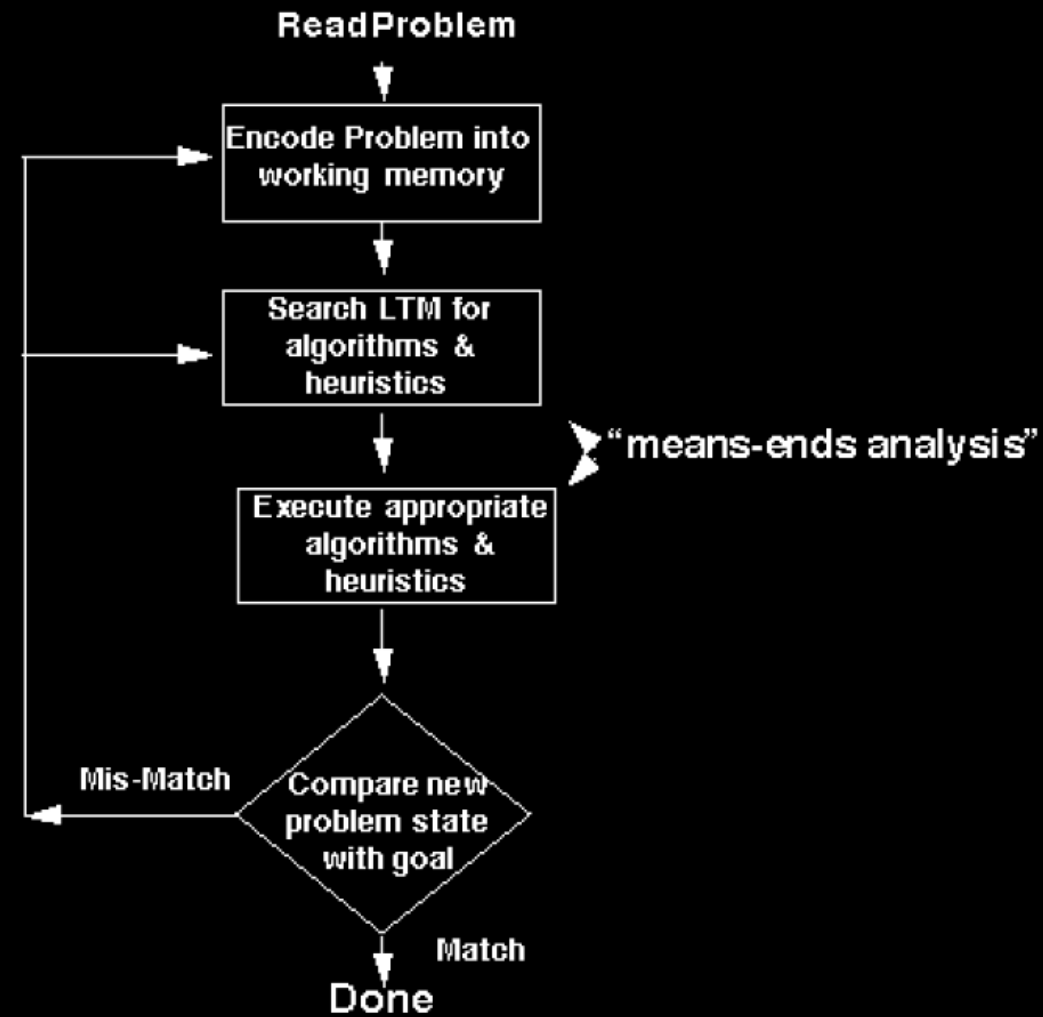
Trait	Novices	Experts
Knowledge	Surface-level, limited	Deep, organized, domain-specific
Strategy	Trial and error	Use of schemas and heuristics
Problem Representation	Focus on surface features	Focus on deep structure of the problem
Speed	Slower, more effortful	Faster, more efficient

Cognitive Framework: Problem Solving Steps

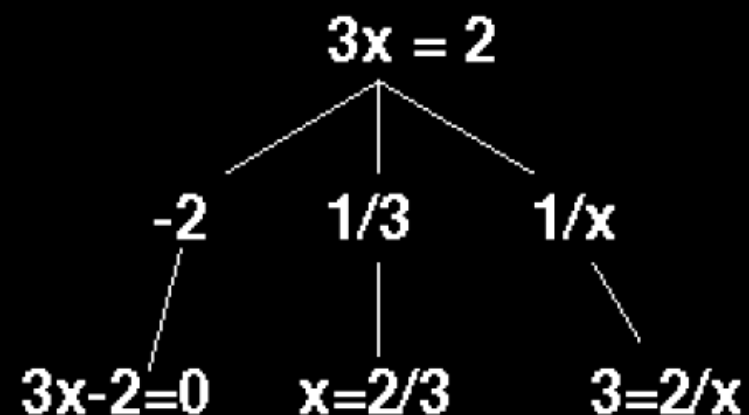
- **Identify the problem** (initial state)
- **Represent the problem** (mentally model it)
- **Choose a strategy** (e.g., means-end analysis, working backward)
- **Apply operators** (actions/moves)
- **Evaluate the solution**
- **Revise if necessary**

Term	Explanation
Problem Solving	Using cognitive processes to reach a goal
Well-Defined Problem	Clear goal, clear steps
Ill-Defined Problem	Ambiguous goal or path
Operators	Actions used to move through problem space
Problem Space	All possible paths between start and goal
Means-End Analysis	Solving by reducing differences via subgoals
Expert vs. Novice	Experts use deeper structures and faster strategies

Based
on
Newell &
Simon
(1972)



Problem Space: a representation of the problem, often in the form of a tree diagram, with possible states and associated operations.



Heuristics: Strategies / "Rules of Thumb"

- **Definition:** Simple, efficient strategies or mental shortcuts used to make decisions and solve problems quickly.
- **Purpose:** They don't guarantee the best solution but often lead to a good-enough solution faster than a full search of all possibilities.

Means-End Analysis

Process:

- Identify the **difference** between your current state and the goal.
- Set a **subgoal** to reduce that difference.
- Choose an **operator** (action/step) that moves you closer to the subgoal.
- **Cycle**: Repeat this until the main goal is reached.

Example:

- **Goal**: Write a research paper.
- **Current state**: You don't have a topic.
- **Subgoal**: Choose a topic.
- **Operator**: Browse articles or brainstorm.
- Once you choose a topic, the next subgoal might be finding sources, and so on.

Anti-Looping Heuristic

- **Purpose:** Avoid going in circles or getting stuck.
- **Rule:** Don't make moves that **increase the distance** from your goal.
- **Example:**
 - In a maze, if a path takes you further from the exit, skip it—even if it looks tempting.
- **In AI or search algorithms**, this is used to **prune** unhelpful options.

Subgoal Structure

- **Definition:** Breaking a problem into **smaller steps (subgoals)** that lead to the final solution.
- **Short-term goals** = Immediate steps (e.g., get books from the library).
- **Long-term goal** = Final destination (e.g., complete your thesis).
- **Example:**
 - Final Goal: Run a marathon.
 - Subgoals: Buy running shoes → Train for 5K → Train for 10K → Increase stamina → Run marathon.

Puzzles

- **Definition:**

Well-defined problems with clear goals, rules, and constraints.

- **Characteristics:**

- Clear **initial state** (starting point)
- Clear **goal state** (what success looks like)
- Defined **rules or operators** that guide moves
- Usually **one correct solution** or a small set of solutions
- Feedback is **immediate** and unambiguous

Examples:

- Sudoku
- Rubik's Cube
- Nine dots puzzle
- Chess puzzles (specific positions)

Cognitive Demands:

- Logical reasoning
- Pattern recognition
- Algorithmic thinking
- Memory for rules and strategies

Real-World Problems

Definition:

Complex, often ill-defined challenges encountered in everyday life or professional settings.

Characteristics:

- Initial state or goal may be **unclear or changing**
- **Rules** may be implicit, flexible, or unknown
- Multiple possible solutions — often no clear “correct” answer
- Feedback can be **delayed**, ambiguous, or conflicting
- Involves multiple factors: social, emotional, economic, etc.

Examples:

- Planning a budget
- Managing a team at work
- Diagnosing a medical condition
- Urban planning or policy making

Cognitive Demands:

- Problem representation and reframing
- Creative and flexible thinking
- Decision making under uncertainty
- Emotional regulation and social cognition
- Long-term planning and monitoring

General Problem Solver (GPS)

- In the GPS theory a person's problem solving efforts are said to take place within a *problem space*, consisting of potential *states* of knowledge and *operators* that transform one state of knowledge into another.
- The **General Problem Solver** (Newell & Simon, 1972) views problem solving as **navigating through a problem space** using **operators** to move from an **initial state** to a **goal state**.

Key concepts:

- **Initial state:** Where the problem begins
- **Goal state:** The desired outcome
- **Operators:** Valid actions or steps to move toward the goal
- **Problem space:** All possible states and transitions
- **Strategies:** Heuristics like means–ends analysis

How Problem Solving Works in GPS:

- **Identify the initial state**
e.g., You're at the start of a puzzle or a real-world task.
- **Define the goal state**
e.g., Solve the puzzle, complete the task, or reach the objective.
- **Explore the problem space**
GPS searches through possible **states and transitions** using **operators**.
- **Apply operators**
These are like "moves" — steps you can take to get closer to the goal.
- **Use strategies** like:
 - **Means–ends analysis**: break the problem into subgoals and reduce the difference between current state and goal.
 - **Working forward or backward**: choose direction of problem solving.

Tower of Hanoi

Rules:

- Only one disk may be moved at a time.
- Only the top disk on a stack may be moved.
- A disk may not be placed on top of a smaller disk.

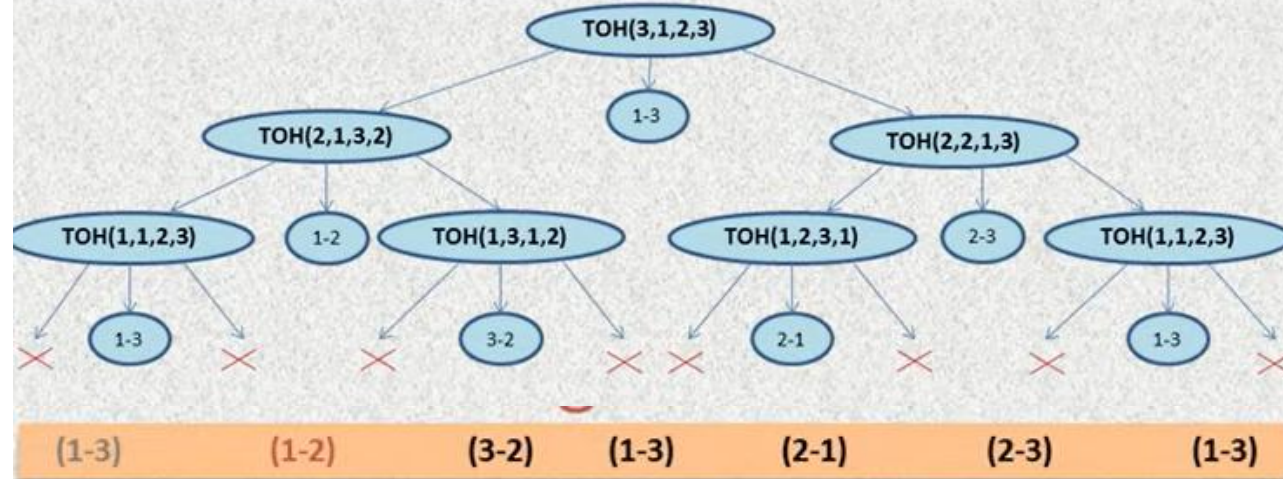
Problem Structure

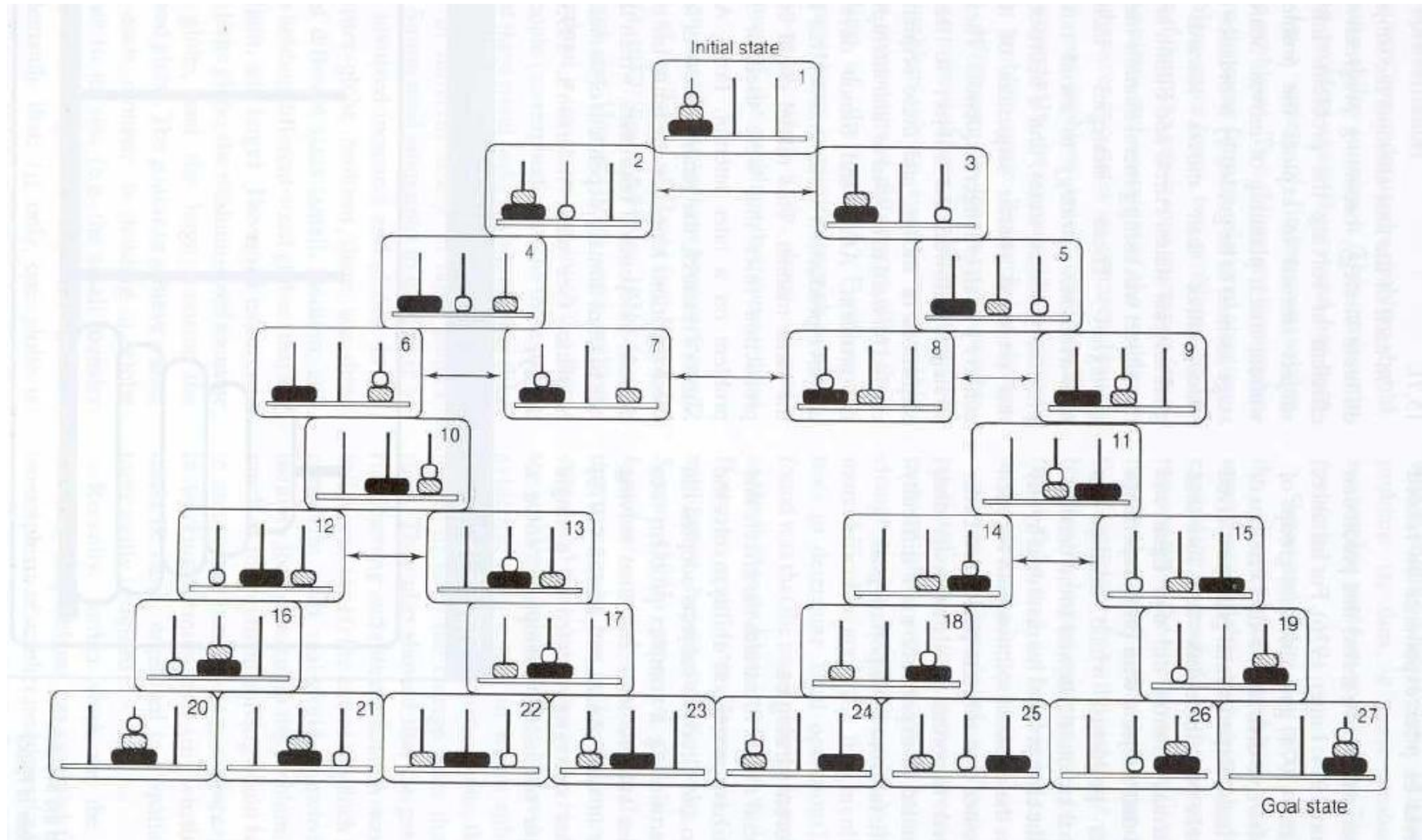
Component	Description
Initial State	All disks stacked on the first peg
Goal State	All disks stacked in order on the third peg
Operators	Move top disk from one peg to another, following rules
Constraints	Only smaller disks can go on larger ones; only move top disk

```

void TOH(int n, int A, int B, int C)
{
    if(n>0)
    {
        TOH(n-1, A, C, B);
        printf( "Move a Disc from %d to %d", A, C);
        TOH(n-1, B, A, C);
    }
}

```





How GPS Approaches Tower of Hanoi

1. Means-End Analysis

- GPS compares current state to goal state.
- Identifies the **difference**: e.g., "Disk 3 is not on peg 3."
- Sets a **subgoal**: "Move Disk 3 to peg 3."

2. Subgoal Structuring

- To move Disk 3, the system sees that smaller disks are in the way.
- New subgoal: Move smaller disks to the auxiliary peg.
- Each subgoal is further broken down until a **primitive operator** (valid move) is executable.

3. Operator Selection

- GPS retrieves **legal moves** from memory (e.g., "Move top disk from peg A to peg B").
- It selects the move that **reduces the difference** between current and goal state.

4. Anti-looping

- GPS avoids repeating the same configurations (e.g., moving disks back and forth endlessly).
- Keeps track of past states or uses rules to prevent backtracking.

Goal: Move Disk 3 to peg 3.

- Subgoal: Move Disks 1 & 2 to peg 2.
 - Subgoal: Move Disk 1 to peg 3.
 - Move Disk 2 to peg 2.
 - Move Disk 1 to peg 2.
- Now move Disk 3 to peg 3.
- Reverse process to move Disks 1 & 2 to peg 3.

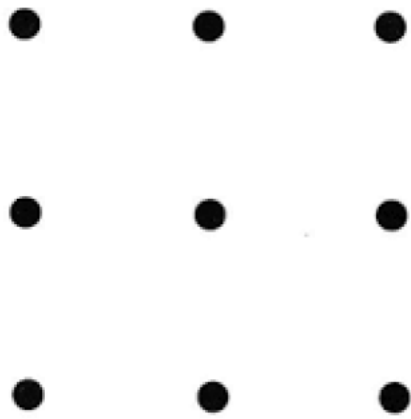
Why Tower of Hanoi is a Good Fit for GPS

- Has a **limited and known problem space**
- Requires clear **subgoal management**
- Can be solved through **recursive decomposition**
- Demonstrates how humans may mentally simulate steps using **heuristics**

Concept	Description
Tower of Hanoi	Puzzle with structured rules and clear goal
GPS	Problem-solving model that uses heuristics like means-end analysis
GPS Solving Hanoi	Breaks the problem into subgoals; chooses operators to reduce difference to goal; avoids loops

Nine dot problem

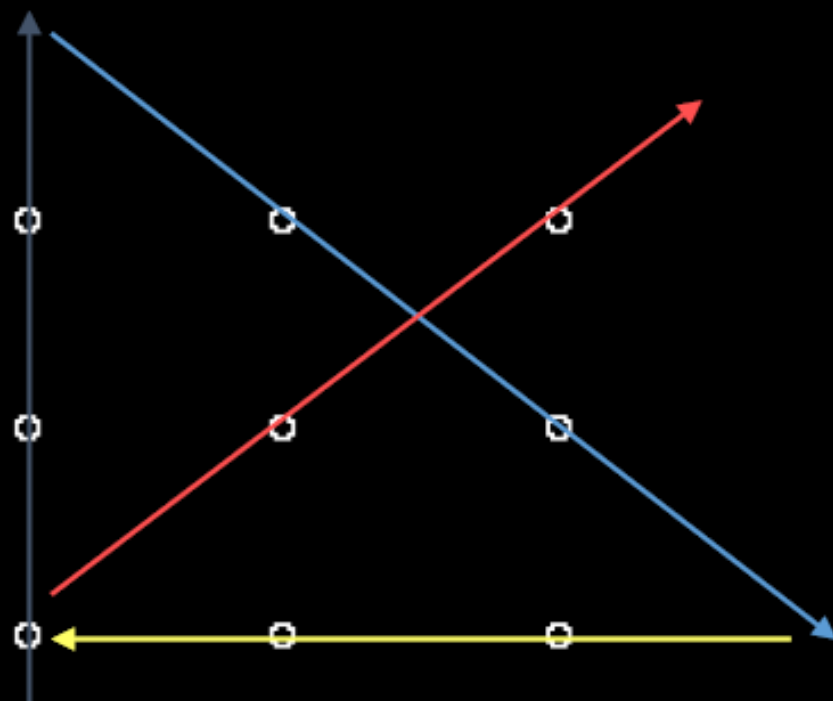
- The nine dots puzzle is a mathematical puzzle whose task is to connect nine squarely arranged points with a pen by four (or fewer) straight lines without lifting the pen.



Rules:

- Connect **all 9 dots** arranged in a 3×3 grid
- Use **four or fewer straight lines**
- **Don't lift your pen**
- Lines can extend beyond the grid boundary

The nine dot problem: connect the dots, with four straight lines, without lifting your pencil or pen.



Solving the 9-Dots Puzzle (Using GPS Concepts)

Element	In the 9-Dots Puzzle
Initial state	Nine unconnected dots
Goal state	All dots connected with ≤ 4 lines
Operators	Drawing a straight line
Constraints	Don't lift pen, max 4 lines
Problem space	All possible line sequences and directions
Solution path	Using 4 extended lines that connect all dots

Assignment-04.09.2025

- The Novice Chess Player as a General Problem Solver
 - Means_ends analysis
- The Beginning Physics Student as a General Problem Solver
 - An object dropped from a balloon descending at 4 meters per second lands on the ground 10 seconds later. What was the altitude of the balloon at the moment the object was dropped?

More Examples

- Water jug problem
- Two-string problem
- Candle Box problem